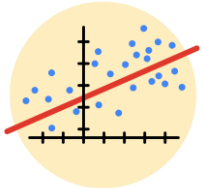
Course Five

Regression Analysis: Simplifying Complex Data Relationships



Instructions

Use this PACE strategy document to record decisions and reflections as you work through this end-of-course project. As a reminder, this document is a resource that you can reference in the future, and a guide to help you consider responses and reflections posed at various points throughout projects.

Course Project Recap

Regardless of which track you have chosen to complete, your goals for this project are:

- Complete the questions in the Course 5 PACE strategy document
- Answer the questions in the Jupyter notebook project file
- Build a multiple linear regression model
- Evaluate the model
- Create an executive summary for team members

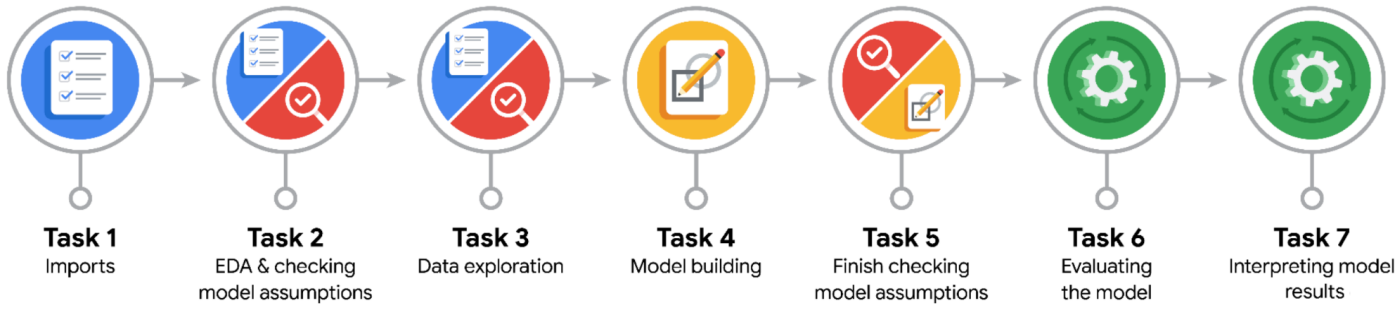
Relevant Interview Questions

Completing the end-of-course project will empower you to respond to the following interview topics:

- Describe the steps you would take to run a regression-based analysis
- List and describe the critical assumptions of linear regression
- What is the primary difference between R^2 and adjusted R^2?
- How do you interpret a Q-Q plot in a linear regression model?
- What is the bias-variance tradeoff? How does it relate to building a multiple linear regression model? Consider variable selection and adjusted R^2.

Reference Guide

This project has seven tasks; the visual below identifies how the stages of PACE are incorporated across those tasks.



Data Project Questions & Considerations

PACE: Plan Stage

- Who are your external stakeholders for this project?

Deshawn Washington the data analysis manager at automatidata and Titus Nelson the operations manager at NYC TLC. Additionally, the data team at automatidata such as Udo Bankole and Uli King.

- What are you trying to solve or accomplish?

To perform a multiple linear regression model in order to predict the fare amount for taxi rides for NYC TLC.

- What are your initial observations when you explore the data?

That there are extreme values in some variables that need to be imputed as these can effect the results.

- What resources do you find yourself using as you complete this stage?

Pandas, NumPy, Seaborn, Matplotlib, statsmodels, datetime and scikit-learn.

PACE: Analyze Stage

- What are some purposes of EDA before constructing a multiple linear regression model?

Some of the purposes of conducting EDA before implementing a multiple linear regression model is to understand the data in relation to the models assumptions. One such example, is the multiple linear regression assumption of linearity, and so through EDA we can see which variables have a linear relationship with each other. Additionally, EDA can be used to clean and prepare the data for modelling.

- Do you have any ethical considerations in this stage?

Yes, I want this model to be ethical and unbiased so that when the predictions are made it is not targeting specific rides or riders. Additionally, I want the model's predictions to be transparent and explainable so that way there is governance in place. Finally, some of the variables have linear relationships with each other as well as being highly correlated with each other.

PACE: Construct Stage

- Do you notice anything odd?

No, the model is improving its predictive performance from the evaluation metrics on the training data to the metrics on the unseen test set. Thereby, proving that the model generalises well to unseen data and has high explanatory power.

- Can you improve it? Is there anything you would change about the model?

Yes, I can improve the model's predictions but I don't need to change the model, only adjust the data. Earlier we found that when the RatecodeID is equal to 2, this is equal to a fare amount of $52, and from further investigating this we found that it is a flat rate fare for JFK International Airport in 2017. Therefore, the predictions from the test set can be adjusted by imputing the fare amount equalling to

$52 when the RatecodeID is equal to 2, and so the model doesn't have to predict these. This was done and the model improved its predictive performance with an increase in the R^2 Score and a decrease in the error metrics.

- What resources do you find yourself using as you complete this stage?

Scikit-learn, pandas, NumPy and statsmodels.

PACE: Execute Stage

- What key insights emerged from your model(s)?

The key insights that emerged from my model is that the residuals of the model follow a normal distribution and are evenly distributed around the zero mark. The coefficients which had the highest weight on the models final predictions was mean_distance with a coefficient of 7.13 and mean_duration with a coefficient of 2.81. Additionally, the intercept of the model showed that the fare for when the variables are at their mean values, is a baseline estimate of $12.89 for a typical trip.

- What business recommendations do you propose based on the models built?

The business recommendations I propose based on the model built is for predictive modelling to be implemented via ensemble learning as this provides more accurate predictions.

- To interpret model results, why is it important to interpret the beta coefficients?

It is important to interpret the beta coefficients because they represent the impact of each independent variable on the dependent variable, while holding other factors constant. Additionally, this helps to identify which variables are the most influential in the model and quantifies how changes in those variables affect the target outcome.

- What potential recommendations would you make?

To do more feature engineering in order to produce more predictive power and to go into predictive modelling with ensemble learners in order to get more accurate predictions. Finally, to adjust the models predictions with domain knowledge such as with the RatecodeID.

- Do you think your model could be improved? Why or why not? How?

Yes, the model can be improved either with more domain knowledge adjustment to the predictions or with new or more features being added or engineered.

- What business/organizational recommendations would you propose based on the models built?

The recommendations that I would propose based on the model built is to implement a hybrid pricing system, where the machine learning predictions are supplemented with predefined pricing rules for known scenarios such as the flat rate fares for JFK airport. By doing this, it ensures both accuracy and consistency with the existing pricing policies. Additionally, I recommend that the model be used for operational decision making, so that the model can be deployed in real time and support the driver and passengers transparency. This also helps to identify any anomalous trips with extremely high or low fares. Finally, I recommend to segment different pricing strategies as the different types of trips, such as airport and regular trips behave differently. This ensures that NYC TLC avoids a one size fits all style approach and instead applies tailored pricing models or rules and treats the segments separately.

- Given what you know about the data and the models you were using, what other questions could you address for the team?

Does the model account for the predefined pricing rules for airport trips?

Can the models predictions be explained?

Can the model generalise well?

Does the model have any bias?

Does the model overfit the data?

- Do you have any ethical considerations at this stage?

Yes, with the main one being that the models predictions are explainable so that they are transparent to the stakeholders.